

COURSE CODE: DSA0603

COURSE NAME: DATA HANDLING AND VISUALIZATION

ASSIGNMENT REPORT

**Design and Evaluation of an Interactive Data Visualization Dashboard for Analyzing
Distributions, Proportions, Relationships, and Patterns Using Perception-Based
Effectiveness Principles**

Dataset: Urban Air Quality & Public Health Visualization Dataset

Team

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Tool: Tableau Desktop Free Edition

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1. Problem Statement and Problem Formulation

1.1 Problem Statement

Public-health organizations need clear and interactive visual analytics to understand environmental conditions that may be associated with population health. Urban air quality is influenced by measurable factors such as particulate matter, nitrogen dioxide, traffic activity, green cover, temperature, and humidity. When these variables are stored only as tables, important distributions, regional differences, trends, and relationships can be difficult to identify.

This project focuses on designing an interactive data visualization dashboard for analyzing an Urban Air Quality & Public Health dataset. The dashboard communicates distributions, proportions, relationships, and temporal or regional patterns using suitable visualization techniques in Tableau. The work applies perception-based effectiveness principles so important information can be identified quickly and compared accurately.

The visualization system demonstrates histograms, box plots, donut/pie charts, bar charts, scatter plots, line charts, and heatmaps. It also evaluates pre-attentive attributes, Gestalt principles, data-ink ratio, color theory, and cognitive load.

1.2 Problem Formulation

The dataset contains 336 records and 16 fields covering 12 Indian cities from 2019 to 2025 across four seasons. It contains categorical variables (City, Region, Season, Air Quality Category), temporal data (Year), and continuous measures including AQI, PM2.5, PM10, NO2, traffic volume, green cover, temperature, humidity, respiratory hospital visits, and respiratory risk index.

Dataset → Data Preparation → Exploratory Analysis → Visualization Selection → Interactive Dashboard → Perception-Based Evaluation → Design Comparison

The main questions are: how is AQI distributed; how does it vary across regions and cities; how are air-quality categories distributed; how does AQI change over time; and what relationships can be explored between AQI and environmental or health-related indicators?

The supplied CSV is a synthetic/educational dataset created for visualization practice. Numerical findings in this report are therefore analysis outputs from the dataset and are not claimed to be official government or clinical statistics.

2. Objective and Expected Outcomes

2.1 Objectives

The main objective is to design and evaluate an interactive visualization dashboard that communicates urban air-quality and public-health-oriented patterns effectively using Tableau and perception-based visualization principles.

1. Understand the structure and quality of the supplied air-quality dataset.
2. Analyze continuous, categorical, and temporal variables using suitable visualizations.
3. Visualize AQI distributions using histograms and box plots.
4. Represent air-quality proportions and city comparisons using donut/pie and bar charts.
5. Identify relationships between AQI, respiratory hospital visits, traffic volume, PM2.5, and green cover.
6. Analyze temporal and regional patterns using line charts and heatmaps.
7. Use color, position, size, and shape appropriately as visual encodings.
8. Evaluate pre-attentive attributes, Gestalt principles, data-ink ratio, color theory, and cognitive load.
9. Compare three alternative visualization designs for the same insight and justify the most effective design.

2.2 Expected Outcomes

The expected outcome is an interactive multi-view Tableau dashboard that enables users to explore air-quality distributions, proportions, relationships, trends, and regional patterns, together with an evidence-based comparison of alternative visual designs.

3. Requirements, Constraints and Assumptions

3.1 Requirements

- Load and organize the supplied CSV dataset.
- Handle categorical, continuous, and temporal variables.
- Provide at least four distinct visualization techniques.
- Include histogram, box plot, pie/donut, bar, scatter, line, and heatmap views.

- Support filters such as Year, Region, City, and Season.
- Use position, color, size, and shape appropriately.
- Provide clear titles, labels, tooltips, and legends.
- Compare three alternative designs for the same insight and rank their effectiveness.

3.2 Constraints

The project is limited to the supplied CSV fields and records. It is an academic visualization prototype, not a clinical or regulatory decision system. Because the dataset is synthetic/educational, observed relationships must not be interpreted as causal public-health evidence.

3.3 Assumptions

- Year is treated as the temporal variable.
- AQI and other numeric fields are treated as continuous measures.
- City, Region, Season, and Air Quality Category are categorical dimensions.
- Average or count aggregation is selected according to the analytical question.
- Users understand that the dataset is intended for educational visualization analysis.

4. Application of Relevant Course Knowledge / Concepts

The project applies core Data Handling and Visualization concepts for transforming tabular data into effective interactive visual information.

- Data handling and field validation.
- Distribution analysis using histograms and box plots.
- Proportion analysis using donut/pie charts.
- Comparison using sorted bar charts.
- Relationship analysis using scatter plots and trend lines.
- Temporal analysis using line charts.
- Pattern analysis using heatmaps.
- Pre-attentive position, color, and size; Gestalt similarity and proximity; data-ink reduction; cognitive-load control.

- Interactive filters, tooltips, and coordinated dashboard views.

5. Design / Proposed Solution / Methodology

5.1 Proposed Solution

The proposed solution is an interactive Tableau dashboard consisting of coordinated analytical views. Each view answers a specific question while sharing common dimensions and filters. The dashboard begins with an overview and supports deeper exploration of distribution, relationships, and regional or temporal patterns.

5.2 Methodology

1. Import the CSV file into Tableau.
2. Review field types and distinguish dimensions from measures.
3. Inspect missing values and data quality.
4. Create calculated/bin fields where required, such as AQI bins.
5. Construct sheets for distribution, proportion, comparison, relationship, trend, and heatmap analyses.
6. Apply visual encodings using position, color, size, and shape.
7. Add filters and tooltips.
8. Combine sheets into dashboards.
9. Evaluate designs using perception and cognitive principles.
10. Compare three alternative designs for one insight and select the most effective.

5.3 System Flow

CSV Dataset → Data Source → Field Validation → Data Preparation → Individual Visualizations → Interactive Dashboard → Perception Evaluation → Alternative Design Comparison → Final Dashboard

6. Algorithm / Pseudocode / Flowchart

6.1 Algorithm

1. Start.

2. Load the CSV dataset into Tableau.
3. Check dimensions, measures, data types, and missing values.
4. Select an analytical question and corresponding variables.
5. Choose a suitable visualization type.
6. Create and format the visualization.
7. Add filters, color, size, labels, and tooltips where useful.
8. Combine related sheets into an interactive dashboard.
9. Evaluate readability using perception-based principles.
10. Create three alternative designs for one selected insight.
11. Rank the designs and justify the final choice.
12. Record results and observations.
13. Stop.

6.2 Pseudocode

BEGIN

Load Dataset

Validate fields and data types

Identify missing values

FOR each analytical question

 Select dimensions and measures

 Select visualization technique

 Build visualization

 Apply position/color/size/shape as appropriate

 Add labels, tooltips and filters

END FOR

Combine selected sheets into dashboards

Evaluate perception, Gestalt, data-ink and cognitive load

Create three alternative designs

Compare designs and select final design

Display dashboard outputs

END

6.3 Flowchart

Start → Load CSV → Validate Data → Prepare Fields → Build Visualizations → Add Interactivity → Evaluate Perception → Compare Designs → Final Dashboard → End

7. Implementation / Source Configuration and Environment / Tools Used

7.1 Implementation

The dashboard is implemented in Tableau Desktop. The CSV file is connected as a text-file data source. Individual worksheets are created for the required analytical views and then assembled into interactive dashboards.

7.2 Main Implementation Modules

Module / Sheet	Function
01 - AQI Distribution	Histogram showing frequency across AQI bins
02 - AQI Box Plot	AQI spread and outliers by region
03 - Air Quality Proportion	Donut/pie chart of category composition
04 - Average AQI by City	Sorted bar chart comparing city averages
05 - AQI vs Respiratory Visits	Scatter plot of AQI and hospital visits
06 - Traffic vs PM2.5	Scatter plot of traffic and particulate matter
07 - AQI Trend by Year	Line chart of temporal trend by region
08 - City Year AQI Heatmap	Heatmap of city-year patterns
09 - Seasonal AQI Comparison	Bar chart comparing seasons
10 - Green Cover vs AQI	Scatter plot of green cover and AQI
11 - Respiratory Risk by City	Bar chart comparing risk index

7.3 Environment and Tools

Component	Tool Used
Visualization Tool	Tableau Desktop Free Edition
Data Source	CSV file
Data Preparation	Tableau data source and worksheet controls
Charting	Show Me, Marks and Analytics features
Interactivity	Filters, tooltips and dashboard actions
Dataset Size	336 rows × 16 fields

7.4 Implementation Flow

CSV connection → field validation → worksheet creation → aggregation selection → visual encoding → formatting → dashboard composition → interaction testing → evaluation.

7.5 Visualization Configuration

AQI bins of size 10 are used for the histogram; average AQI is used for regional and city comparisons; count of records is used for category proportions; Year is used for the temporal axis; scatter plots use positional encodings and can use Region as color and Traffic Volume Index as size.

8. Test Cases and Expected / Actual Results

8.1 Test Cases

Test Case	Expected Result	Observed Dataset Output
AQI distribution	Histogram shows frequency across AQI ranges	AQI ranges from 59.77 to 223.12.
Regional AQI	Box plots show spread and central tendency	North mean AQI 157.54; South 121.14.
Category proportion	Donut/pie shows composition	Unhealthy 283; Good/Moderate 45; Very Unhealthy 8.
City comparison	Bars identify higher/lower cities	Delhi 180.51 mean AQI; Chennai 115.13.
AQI-health relationship	Scatter reveals association patterns	Positive linear association in the supplied dataset.
Temporal trend	Line chart reveals yearly movement	Mean AQI 143.74 in 2019 and 134.49 in 2025.
City-year pattern	Heatmap exposes differences	Delhi remains among highest-AQI cities in the dataset.

8.2 Data Quality Observation

The dataset contains 336 records. PM10 has 10 missing values, Green Cover has 10 missing values, and Humidity has 10 missing values. Missing values should not be silently treated as zeros.

9. Execution Screenshots / Outputs

The following figures are analytical output representations generated from the supplied dataset and can be reproduced in Tableau using the configurations in Section 7.

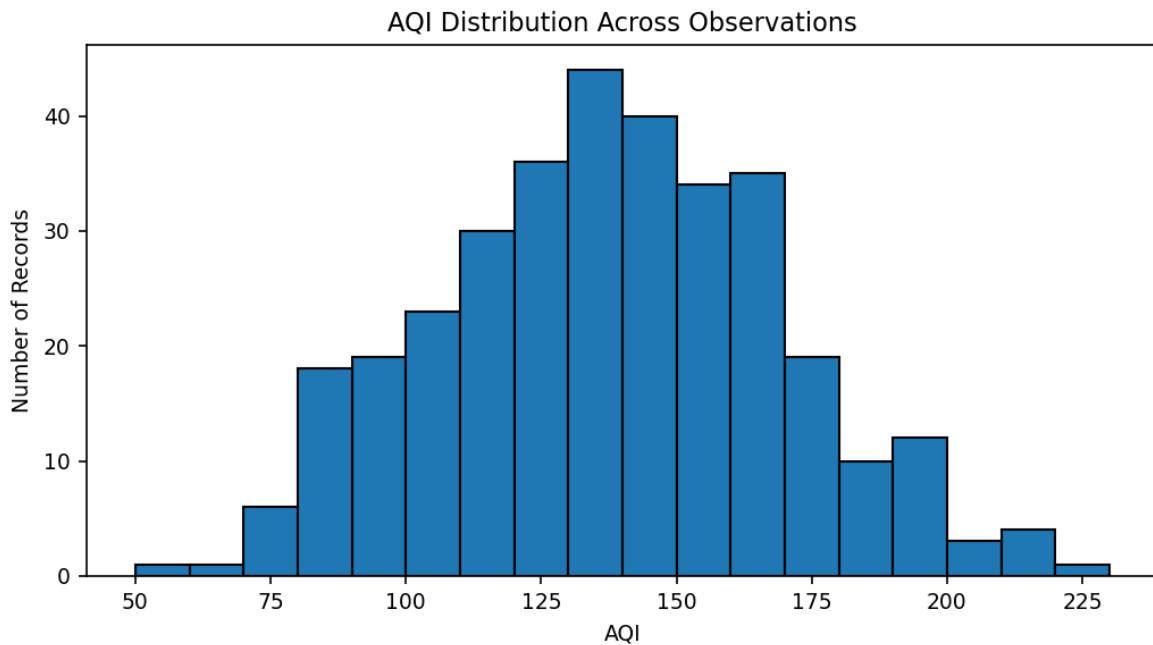


Figure 9.1 – AQI Distribution Across Observations

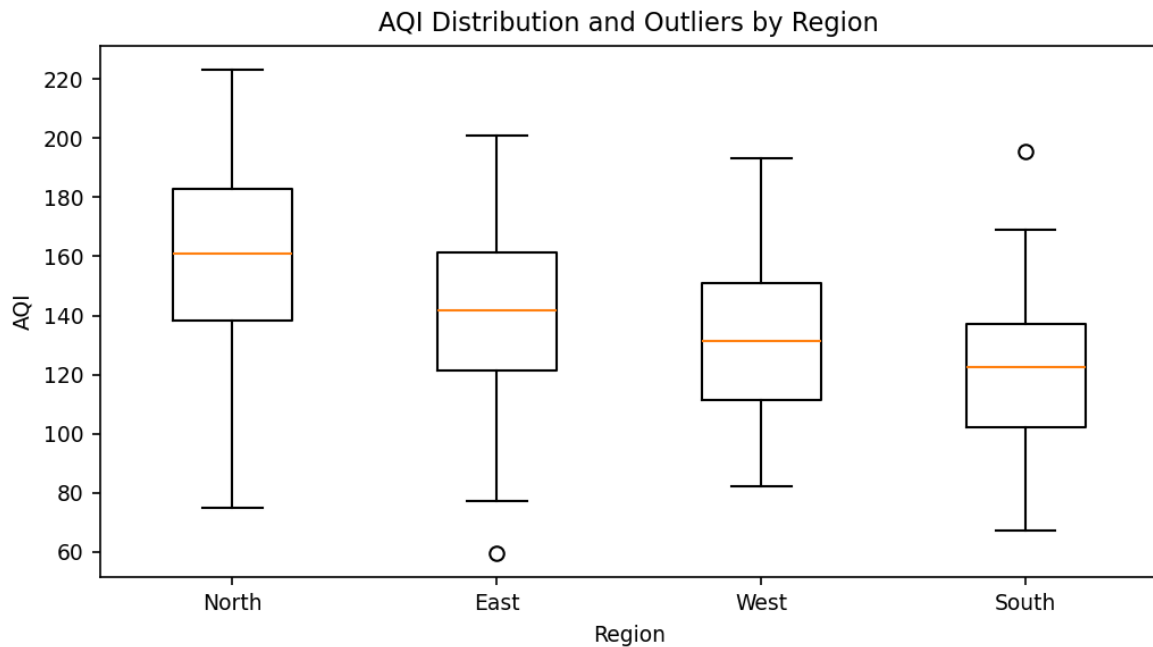


Figure 9.2 – AQI Distribution and Outliers by Region

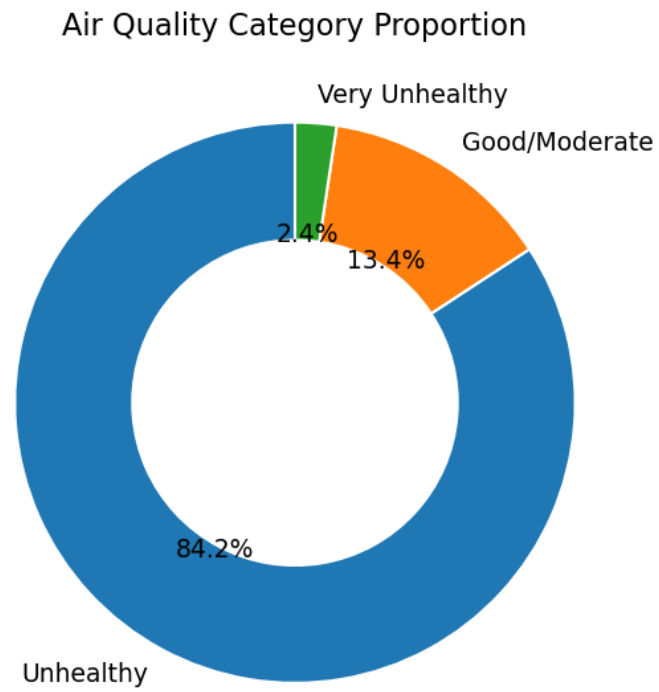


Figure 9.3 – Air Quality Category Proportion

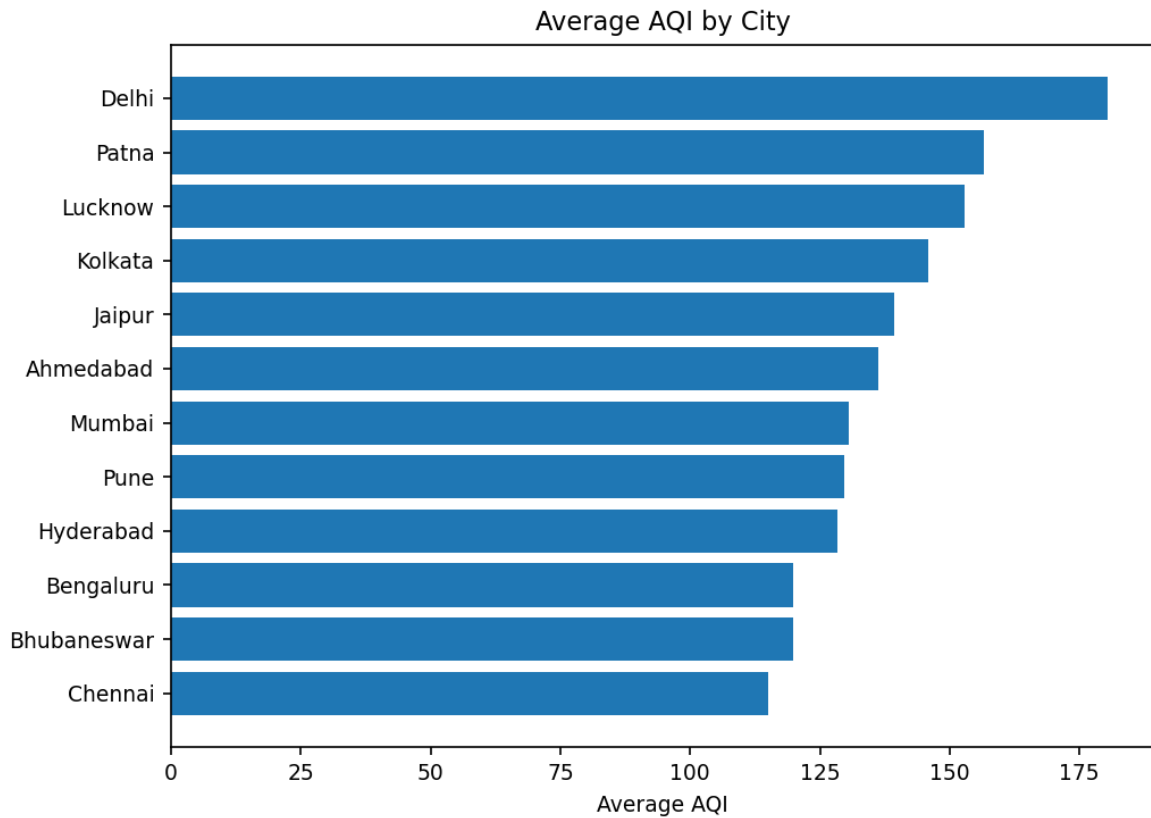


Figure 9.4 – Average AQI by City

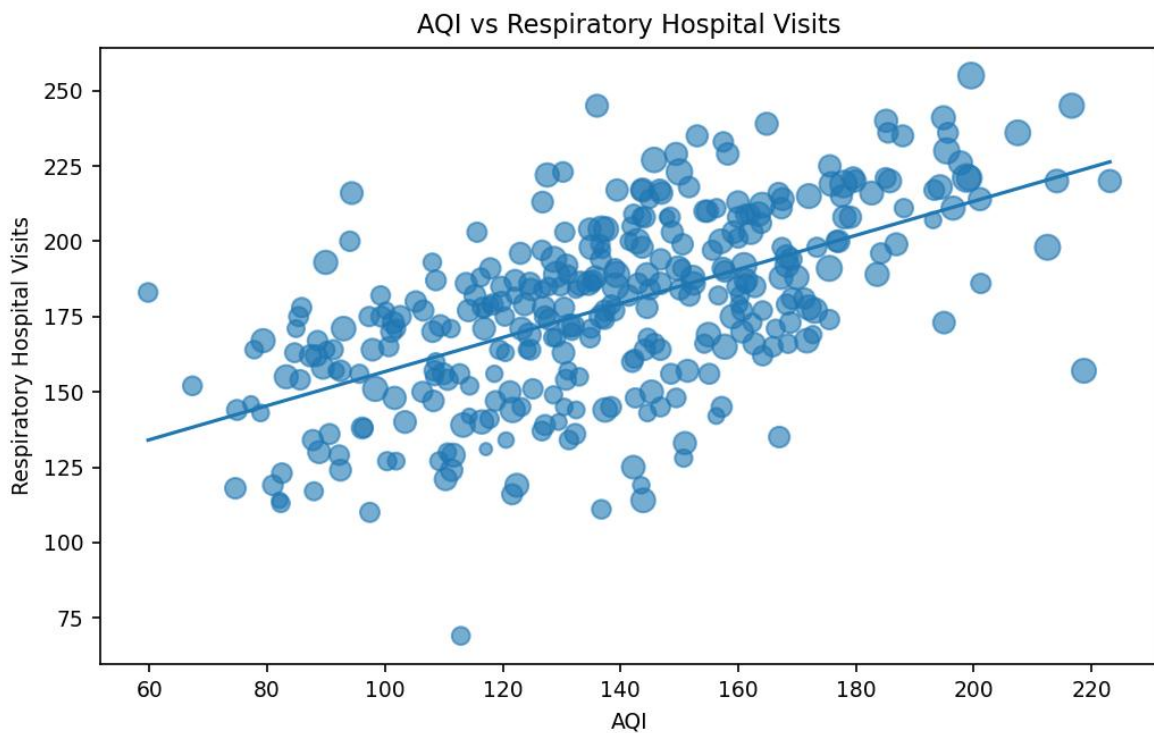


Figure 9.5 – AQI vs Respiratory Hospital Visits

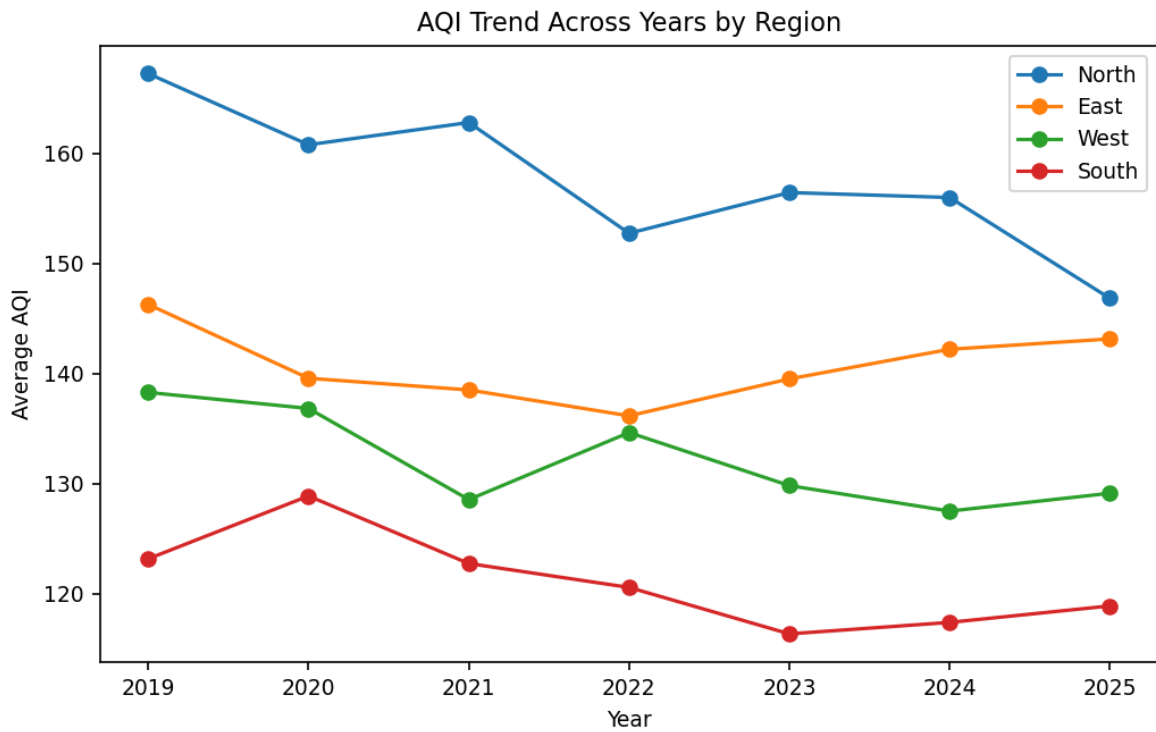


Figure 9.6 – AQI Trend Across Years by Region

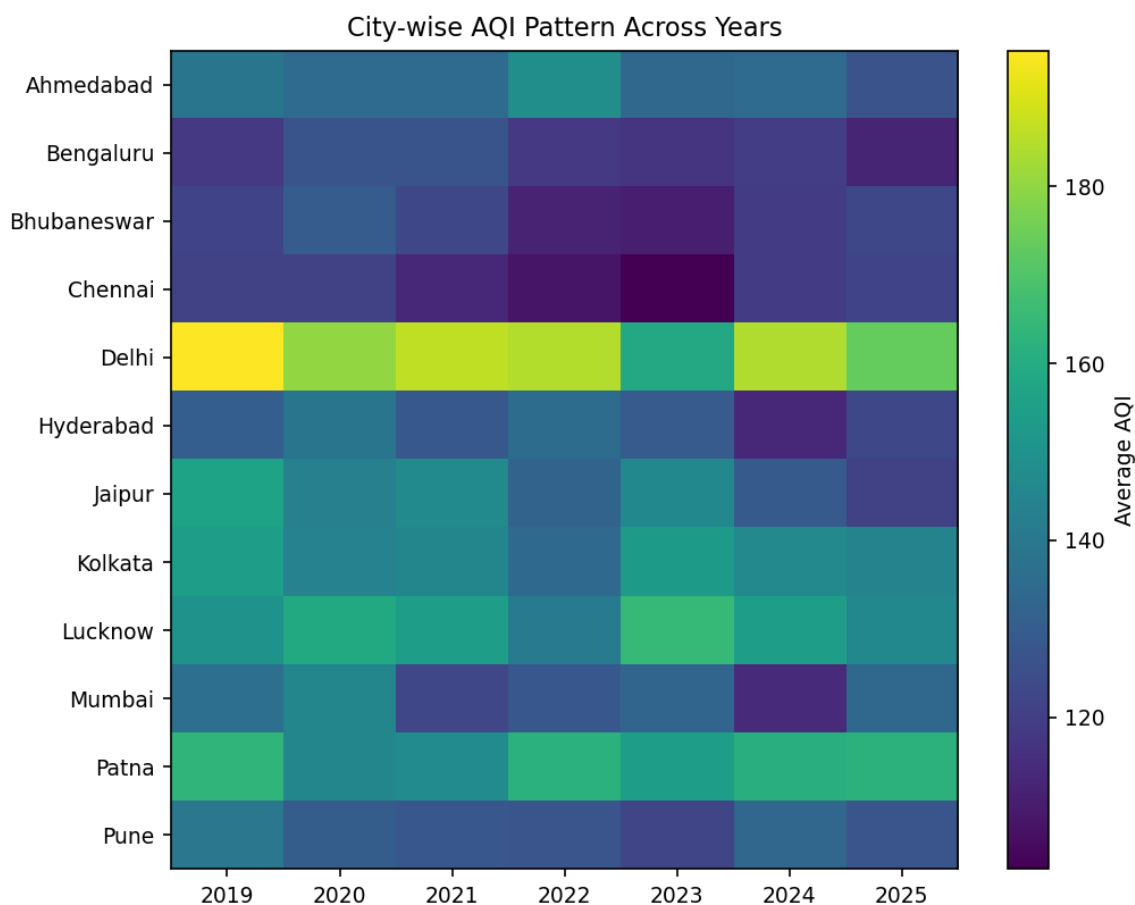


Figure 9.7 – City-wise AQI Pattern Across Years

10. Results and Validation

The overall average AQI is 137.88, with values ranging from 59.77 to 223.12. The category distribution is concentrated in the Unhealthy class, which accounts for 283 of the 336 records.

10.1 Validation Results

Parameter	Result
Records analyzed	336
Fields analyzed	16
Cities	12
Years	2019–2025
Seasons	4
Average AQI	137.88
Minimum AQI	59.77
Maximum AQI	223.12

Highest mean regional AQI	North – 157.54
Lowest mean regional AQI	South – 121.14
Highest mean city AQI	Delhi – 180.51
Lowest mean city AQI	Chennai – 115.13

10.2 Experimental Observation

Yearly mean AQI declines overall from 143.74 in 2019 to 134.49 in 2025, although the trend is not perfectly monotonic. Regional averages also differ substantially.

10.3 Visualization Validation

Histogram, box plot, donut/pie, bar, scatter, line, and heatmap views map the intended variable types appropriately and together cover distributions, proportions, comparisons, relationships, time, and two-dimensional patterns.

10.4 Evidence

Figures 9.1–9.7 provide evidence for the major analyses. Values are calculated directly from the supplied CSV and are not official public-health statistics.

11. Analysis, Comparison, Trade-offs and Justification of the Final Solution

11.1 Analysis

The analysis shows substantial city and regional variation. Delhi has the highest mean AQI at 180.51 and Chennai the lowest at 115.13. North has the highest regional average at 157.54, followed by East 140.76, West 132.10, and South 121.14. The category distribution is dominated by Unhealthy observations.

The AQI versus respiratory hospital visits view is useful for pattern exploration because both measures share common axes, while bubble size can represent traffic volume. This is an association analysis and does not establish causation.

11.2 Comparison of Three Alternative Designs

Selected insight: How are air-quality categories distributed across the dataset? The same category-count information was represented using a bar chart, donut/pie chart, and stacked bar chart.

Criterion	Design 1: Bar	Design 2: Donut/Pie	Design 3: Stacked Bar
Comparison accuracy	5/5	3/5	4/5
Ranking categories	5/5	2/5	4/5
Ease of interpretation	5/5	4/5	3/5
Pattern detection	4/5	3/5	4/5
Cognitive load	Low	Medium	Medium
Decision usefulness	5/5	3/5	4/5
Overall rank	1st	3rd	2nd

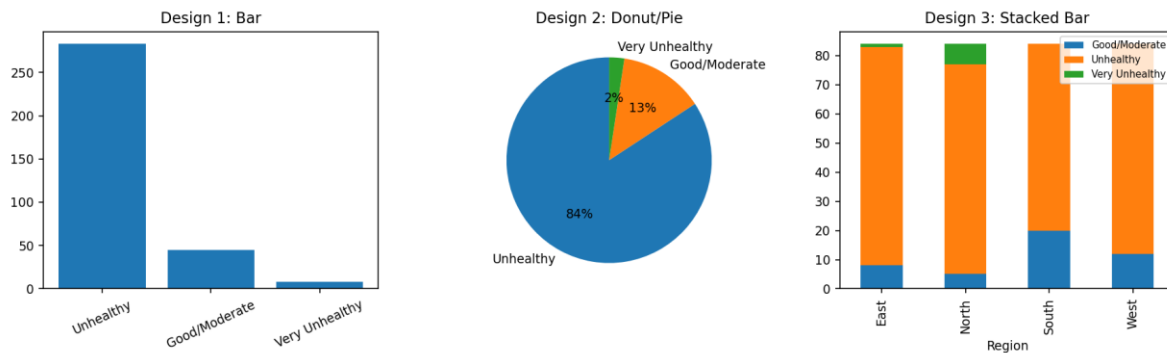


Figure 11.1 – Three Alternative Designs for the Same Insight

11.3 Trade-offs

The bar chart provides the strongest quantitative comparison because position and length are easy to compare against a common baseline. The donut/pie chart communicates part-to-whole composition but makes precise comparison harder. The stacked bar provides regional context but can increase comparison difficulty when segments lack a common baseline.

11.4 Justification of the Final Solution

The bar chart is selected as the most effective primary design for category comparison because it minimizes cognitive effort and supports direct ranking. The donut is retained as a secondary overview view for composition. Multiple coordinated views are used instead of relying on a single chart type.

12. Broader Considerations / SDG Relevance

12.1 Broader Considerations

Air-quality visualization can help users explore where and when higher AQI values occur. In an actual public-health setting, operational use would require validated measurements, documented provenance, privacy safeguards for health data, and domain-expert review.

12.2 SDG Relevance

The project is relevant to SDG 9: Industry, Innovation and Infrastructure through digital analytics and visualization. It is also thematically connected to SDG 3: Good Health and Well-being through environmental and respiratory-health-oriented analysis, and to SDG 4: Quality Education through the development of data-literacy and visualization skills.

13. Conclusion, Limitations and Possible Improvements

13.1 Conclusion

The project demonstrates an interactive visualization system for urban air-quality and public-health-oriented analysis. Multiple visualization techniques communicate distributions, proportions, comparisons, relationships, trends, and patterns. Perception-based principles support effective visual communication, while the design comparison identifies the bar chart as the strongest choice for accurate category ranking.

13.2 Limitations

- The supplied dataset is synthetic/educational rather than an official measured dataset.
- The dataset covers a fixed set of cities, years, and seasons.
- The analysis identifies associations but does not establish causal relationships.
- Missing values exist in PM10, Green Cover, and Humidity.
- The dashboard is an academic prototype and is not intended for clinical or regulatory decisions.
- Visualization effectiveness can vary with audience, screen size, and interaction context.

13.3 Possible Improvements

1. Replace the educational dataset with a validated open government or public-health dataset for real-world deployment.
2. Add more cities, geographic levels, and longer time periods.
3. Add geographic maps and spatial filtering.
4. Add data-quality indicators and preprocessing documentation.
5. Add parameter controls for metric selection and threshold analysis.
6. Conduct a user evaluation using task completion time and interpretation accuracy.
7. Improve accessibility using color-blind-safe palettes and alternative encodings.

14. Individual Contribution of Group Members

The project work was divided among the three group members, with all members participating in discussions, testing, dashboard refinement, and final report preparation.

Group Member	Contribution
Ch. Tharunika Reddy	Dataset understanding, data preparation, and distribution/proportion visualizations.
R Anu	Relationship, trend, regional visualizations, and dashboard arrangement.
Pradeep Kumar S	Perception-based evaluation, design comparison, validation, result analysis, and report preparation.

15. References

1. Tableau, Tableau Desktop and Tableau visualization documentation.
2. Python Software Foundation, Python Documentation — supporting data-analysis concepts and dataset inspection.
3. DSA0601 – Data Handling and Visualization assignment instructions supplied for the project.
4. United Nations, Sustainable Development Goals.
5. Supplied dataset: Urban_Air_Quality_Public_Health_Visualization_Dataset.csv (educational/synthetic dataset created for visualization practice).